

The George Washington University
Dept of Electrical and Computer Engineering

ECE 6213
Fall 2026
Dr. Matthew LaRue

Project 3: Elevator Controller FSM

Academic integrity expectation: Students will submit project in groups of two, submitted project should be group's own work. Groups can collaborate for help with conceptual understanding of project or to troubleshoot coding and simulation issues. Only one group member should submit project to blackboard, and both group member's names should be on the submission.

Design an elevator controller FSM for a three story elevator. Required inputs/outputs are listed below. The FSM will interface with an elevator button Verilog block that I will provide.

Inputs:

Clock, reset_n
Floor 1 up button
Floor 2 down button
Floor 2 up button
Floor 3 down button
Elevator floor 1 button
Elevator floor 2 button
Elevator floor 3 button

Outputs:

Floor 1 (indicates elevator is on floor 1)
Floor 2
Floor 3
Elevator door open
Floor 1 up button clear (explanation for all the "clear" signals are below)
Floor 2 down button clear
Floor 2 up button clear
Floor 3 down button clear
Elevator floor 1 button clear
Elevator floor 2 button clear
Elevator floor 3 button clear

The overall functional requirement is that it functions like a normal three story elevator. See test cases at the bottom to help understand the functionality.

The only sequential variable allowed is the current FSM state. All other logic must be combinational logic using the current inputs and current state. For example, you cannot

create sequential signals such as “destination floor” or “traveling up/down” that you use to store this information. Instead, you must encode this information into the FSM states

All 7 elevator button inputs to your FSM will connect to the provided `elevator_button.v` module. When you press the button to call an elevator, that button always stays lit until the elevator as completed request (ex. If you are on floor 1 of a building a press the up botton, that button will stay lit until the elevator appears at your floor and allows you to enter). The functional waveforms for the module are shown in figure 1, and figure 2 shows how you should implement this module in the design hierarchy. The “button pressed” input should come from your testbench, and then then “clear” signal should be generated by your FSM.

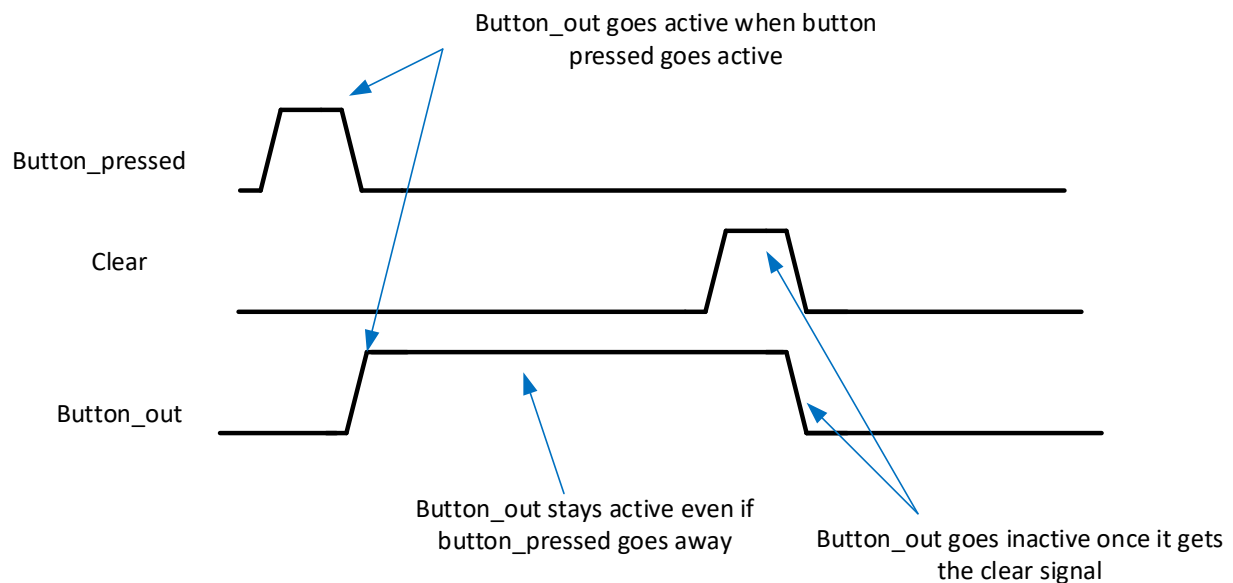


Figure 1. Elevator button module waveforms

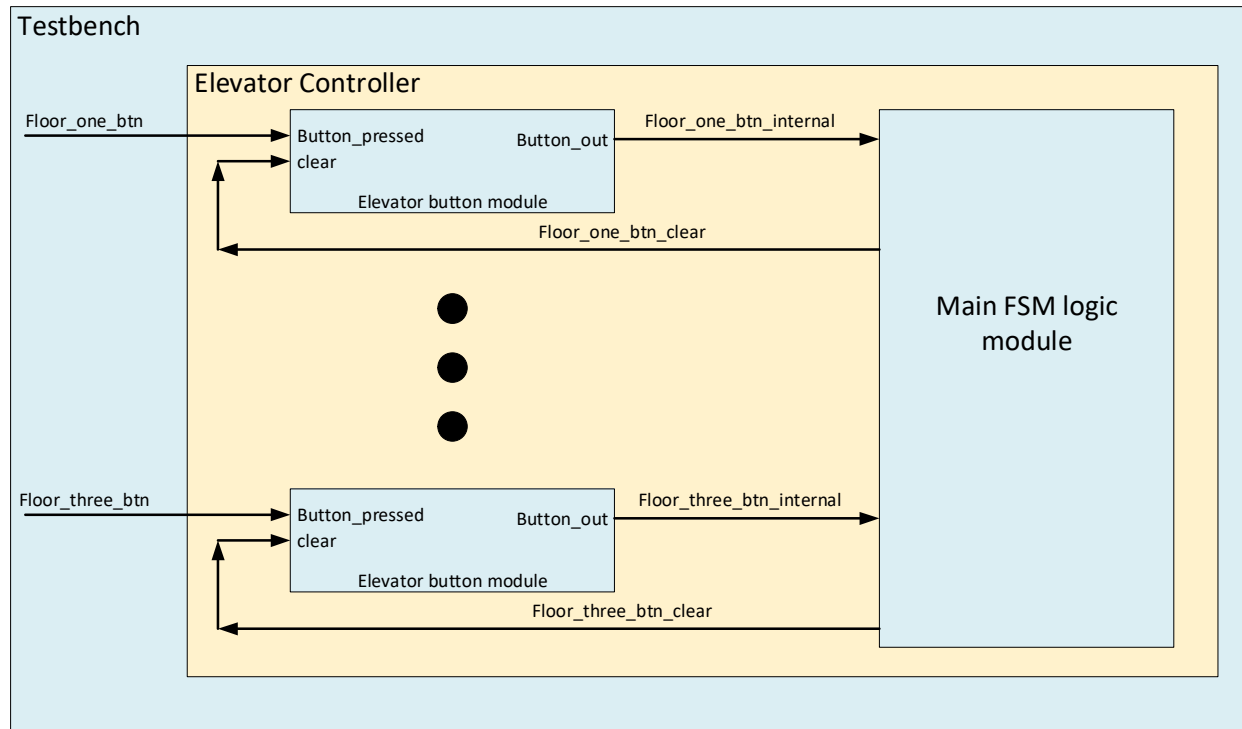


Figure 2. Design hierarchy

Deliverable 1: Due Friday, September 18

Worth 50 points

Provide the initial state transition diagram for your elevator FSM.

Deliverable 2: Due Friday, September 25

Worth 150 points

1. Elevator FSM Verilog code
2. Testbench that *fully* demonstrates elevator functionality
3. Simulation waveforms demonstrating all test cases listed below
 - a. Make sure the waveforms are legible, and label the waveforms so I know which test case is being demonstrated
4. Provide the final state transition diagram for your elevator FSM.
 - a. If the final FSM is different from your deliverable 1 FSM, provide details on what changed and why (ie. simulations showed a testcase that didn't operate properly)

Below are a list of the specific test cases that you should simulate in your elevator controller FSM testbench. These testcases can also be helpful as you are designing your FMS because they are decent examples of "plain English descriptions" of the FMS functionality.

Test Cases:

1. Floor 1 -> Floor 2: Passenger hits "floor 1 up" button. Elevator opens doors on floor one and "floor 1 up" button clears. The door closes and the passenger hits the "floor 2" button. The elevator travels to floor two, opens its doors, and clears the "floor 2" button
2. Floor 1 -> Floor 3: Passenger hits "floor 1 up" button. Elevator opens doors on floor one and "floor 1 up" button clears. The door closes and the passenger hits the "floor 3" button. The elevator travels to floor two and keeps its doors closed. It then proceeds to floor three, opens its doors, and clears the "floor 3" button
3. Floor 2 -> Floor 3: Passenger hits "floor 2 up" button. Elevator opens doors on floor two and "floor 2 up" button clears. The door closes and the passenger hits the "floor 3" button. The elevator travels to floor three, opens its doors, and clears the "floor 3" button
4. Floor 2 -> Floor 1: Passenger hits "floor 2 down" button. Elevator opens doors on floor two and "floor 2 down" button clears. The door closes and the passenger hits the "floor 1" button. The elevator travels to floor one, opens its doors, and clears the "floor 1" button
5. Floor 3 -> Floor 2: Passenger hits "floor 3 down" button. Elevator opens doors on floor three and "floor 3 down" button clears. The door closes and the passenger hits the "floor 2" button. The elevator travels to floor two, opens its doors, and clears the "floor 2" button
6. Floor 3 -> Floor 1: Passenger hits "floor3 down" button. Elevator opens doors on floor three and "floor 3 down" button clears. The door closes and the passenger hits the "floor 1" button. The elevator travels to floor two and keeps its doors closed. It then proceeds to floor one, opens its doors, and clears the "floor 1" button
7. Floor 1 -> Floor 3 w/ pickup on Floor 2: Passenger hits "floor 1 up" button. Elevator opens doors on floor one and "floor 1 up" button clears. The door closes and the passenger hits the "floor 3" button. There is also a passenger on floor two that hits the "floor 2 up" button. The elevator travels to floor two and opens its doors. The passenger gets on, the "floor 2 up" button clears, and the doors close. It then proceeds to floor three, opens its doors, and clears the "floor 3" button.
8. Floor 3 -> Floor 1 w/ pickup on Floor 2: Passenger hits "floor 3 down" button. Elevator opens doors on floor three and "floor 3 down" button clears. The door closes and the passenger hits the "floor 1" button. There is also a passenger on floor two that hits the "floor 2 down" button. The elevator travels to floor two and opens its doors. The passenger gets on, the "floor 2 up" button clears, and the doors close. It then proceeds to floor one, opens its doors, and clears the "floor 1" button
9. No passengers stuck: If a passenger inside the elevator hits any of the "floor 1", "floor 2", or "floor 3" buttons, the elevator should eventually get to that floor and let the passenger out. For example with test case #2, if the passenger that gets in on floor two hits the "floor 1" button (even though the elevator is going up), after dropping off the first passenger on floor three the elevator should travel from floor three to floor one as described in test case #6.